Supply Chain SLA Diagnostic & Fulfillment Bottleneck Analysis

Project Summary

Core Business Question: *"Where in the fulfillment lifecycle are delivery delays originating—seller dispatch or courier transit—and which specific shipping corridors and product categories systematically breach promised delivery SLAs?"*

1. Executive Overview & Key Findings

Across 110,197 completed order line items analyzed from the dataset, the enterprise recorded an overall Customer Delivery SLA Breach Rate of 7.91% Root-cause decomposition proves conclusively that delivery delays are primarily a carrier-side breakdown rather than an upstream merchant fulfillment problem.

- **Root Cause Attribution:** 71.63% of all SLA breaches are strictly from Carrier Transit Bottlenecks where the merchant handed over packages on schedule, but road transport missed the customer estimated delivery date. Seller dispatch delays accounted for only 28.37%.
- **Geographic Fragility:** Cross-state (Inter-State) deliveries suffer a 9.00% breach rate—50% higher than intra-state movements (5.98%).
- **Route Corridor:** SP -> MA shipping lane exhibits an acute 22.22% breach rate—more than 1 in 5 orders arrive late. SP -> CE (15.40%), SP -> RJ (14.85%), and SP -> BA (14.78%) follow closely.
- **Category & Weight Vulnerabilities:** High-value, bulky, and fragile categories lead delivery failures (Home Comfort at 10.26%). Packages over 10 kg surge to a 10.61% breach rate (a 40.9% jump compared to under 1kg parcels at 7.53%).

2. Executive KPI Scorecard

Metric Name	Value	Operational Definition & Impact
Total Shipments	**110.197K**	Total completed line items delivered across platform (2016–2018)
Customer SLA Breach Rate	**7.91%**	8,722 shipments delivered past promised customer estimated date
Seller Dispatch Breach Rate	**9.31%**	Orders handed off past merchant shipping limit date (First-mile delay)
Carrier Failure Rate	**5.66%**	On-time merchant dispatch that subsequently breached during road transit
Avg Carrier Transit Days	**9.19 Days**	Mean road transit time from carrier pickup to customer doorstep
Inter-State Breach Gap	**+50.5% Gap**	9.00% (Inter-State) vs 5.98% (Intra-State baseline)
Worst Corridor Breach	**22.22% (SP->MA)**	Leading route corridor failure rate

3. Statistical Thresholds & Sample Integrity

In operational analytics, unconstrained aggregations create deceptive outlier traps where remote routes or rare products with 2 orders show 100% false-alarm breach rates. Objective volume thresholds were engineered to insulate strategic decisions:

- **Product Categories (>= 300 Orders | ~95.3% Volume Coverage):** Filters out 38 long-tail niche categories carrying under 5% combined volume, isolating 34 core categories.
- **Shipping Corridors (>= 500 Shipments | 86.2% Trunk Coverage):** Brazil comprises 400+ routes. A 500-order cutoff isolates 30 major lines where daily scheduled operates (~1 dispatch/day over Olist's 20-month horizon).

4. Strategic Action Plan & Priority Roadmap

- **Dynamic SLA Lead Times:** Add a 3–4day dynamic buffer on routes (`SP -> MA`, `SP -> CE`, `SP -> BA`) at checkout to match real transit capacity.
- **Dedicated Heavy weight carriers:** Contract dedicated carriers for bulky goods (**>10 kg**) to fix **10.61% breach rate.
- **Regional Warehousing:** Move more volume product categories closer to regional hubs, shifting high-risk inter-state routes (**9.00%**) into local intra-state deliveries (**5.98%**).
- **Carrier Performance Penalties:** Enforce SLA penalty clauses for carriers failing an 8% breach threshold.

5. Tech Stack

• MySQL: Data ingestion via `LOAD DATA LOCAL INFILE`, relational joins, and engineered the analytical view `delivery_sla_analysis`.

• Power BI: Built 2-page executive report with custom KPIs, visuals and optimized DAX measures.